

Temporal Trends in Psychometric Reporting Quality in Psychology

Candidate Number: 307351

Word Count: 7000

Date of Submission: 27/08/2026

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Acknowledgements

I am grateful to my supervisor, Professor Andy Field, for his guidance, patience, and support throughout this project. I would also like to thank Amelia, whose help as a fellow researcher was invaluable, and Dr Jenny Terry, for her advice and assistance along the way. My thanks go to the team of coders whose careful work made the data collection for this study possible.

I am thankful to my friends and family for their encouragement and support throughout. Finally, I would like to acknowledge the artists whose music kept me company through the long hours of analysis and writing.

Abstract

Sound measurement is fundamental to psychological science, yet the transparency with which psychometric instruments are developed, validated, and reported is unclear. Although the past two decades have seen the publication of formal reporting standards and a broad movement toward methodological reform following the replication crisis, whether these developments have improved the reporting of newly developed and modified measures remains unknown. The present study examined whether the quality of psychometric reporting changed across the period 2002 to 2026, using a large sample of 1,142 measures nested within 530 articles drawn from five psychology subfields. Reporting quality was operationalised as the proportion of preregistered coding criteria met for each measure, and modelled using a Bayesian multilevel beta regression with a piecewise linear effect of publication year, incorporating a breakpoint fixed at 2014 to capture any shift associated with the revised reporting standards and the reform period that followed. Reporting quality did not credibly change over the study window, either overall or within any of four sub-dimensions comprising scale construction, sample description, validity and reliability, and dimensionality, and no credible change occurred at the 2014 breakpoint. Reporting was uniformly low and differed markedly across sub-dimensions, from sample description at the highest level to validity and reliability evidence at the lowest, though this ordering remained stable throughout. Variation in reporting quality was attributable to differences between articles rather than to publication year. These findings indicate that methodological reform has not reached the reporting of measurement, and suggest that structural, enforced interventions are now required.

Temporal Trends in Psychometric Reporting Quality in Psychology

A fundamental aspect of psychological science is defining and measuring what is being studied. Before the aetiology of depression can be investigated or the efficacy of a treatment evaluated, the construct must be defined and instruments capable of measuring it accurately must be built (Flake & Fried, 2020). This is difficult because constructs such as depression, personality traits, attitudes, and cognitive abilities rarely permit direct observation. Construct validity, the degree to which a measure's scores track a person's true standing on the underlying construct, is therefore foundational to psychological measurement (Clark & Watson, 2019; Cronbach & Meehl, 1955). Because it can only be judged when authors report sufficient detail about their measures, incomplete reporting undermines both the assessment of construct validity and the interpretability of research claims (Lawson et al., 2026). When construct validity is poor, scores do not reflect the attribute they are claimed to, and the resulting inferences are compromised in a way that no amount of rigorous design, advanced statistics, or large samples can correct (Flake & Fried, 2020; Loevinger, 1957). The quality with which measurement is conducted, and the transparency with which it is reported, is thus not one methodological concern among many but a precondition for the accumulation of psychological knowledge.

The state of psychometric reporting

Despite the foundational role measurement plays in the validity of study conclusions, important information about measurement is routinely absent from published psychology articles. Measures are used without reference to their source, evidence that they capture the intended construct is frequently lacking, and unjustified measurement flexibility abounds. Barry et al. (2014) reported that between 40% and 93% of measures used across seven journals in health education lacked validity evidence, and Weidman et al. (2017) found that 69% of 356

measurement instances in emotion research included no reference to prior research or any systematic development process. In a representative sample, Flake et al. (2017) reported that 40% of measures gave no indication of their source, 19% omitted the number of items, and 9% the response scale, and that where reliability was reported it was very often Cronbach's alpha alone.

The problem deepens when reporting is examined not as a matter of what is disclosed but of what is true. Hussey and Hughes (2020) subjected 15 widely used self-report measures to comprehensive validity assessment, including factor structure, test-retest reliability, and measurement invariance. Only 4% demonstrated good validity, though 88% appeared adequate on internal consistency alone. The authors termed this hidden invalidity, warning that the underreporting of structural validity may conceal an abundance of invalid measures hiding in plain sight. The implication is pointed: the one property psychologists report with any regularity is close to useless as a guide to whether a measure is valid. A subsequent review reinforced the point, finding that only about 9% of multi-item scales in the studies underlying the Reproducibility Project were accompanied by any factor analysis (Flake et al., 2022).

These concerns are not confined to self-report questionnaires. Parsons et al. (2019) argued that psychology has not developed a tradition of routinely examining or reporting the reliability of cognitive and behavioural task measures, and that establishing such a practice is a precondition for drawing robust inferences from the analyses they support. That the difficulty has become a recognised area of study is itself telling: Meier (2023) framed an entire collection around the persistence of measurement problems, signalling a stubborn and ongoing feature of contemporary practice. Together, this work establishes that inadequate psychometric reporting is broad and persistent, spanning subfields, measure types, and categories of validity evidence.

What it does not establish is whether the situation has changed over time, since almost every study offers a cross-sectional snapshot rather than asking whether reporting quality has improved, stagnated, or deteriorated.

Questionable measurement practices and researcher degrees of freedom

Flake and Fried (2020) introduced the term "questionable measurement practices" for the decisions that raise doubts about the validity of the measures used in a study, and ultimately about the validity of its conclusions. Measurement, they argued, comes with a garden of forking paths: data from a ten-item questionnaire can be summarised into a single sum score in more than a thousand ways, and the possibilities multiply once subscales and alternative analyses are considered. This flexibility exists whether or not a researcher intends to exploit it, and when the decisions taken are neither reported nor justified, the scientific community cannot evaluate the study's internal, external, statistical-conclusion, and construct validity. Questionable measurement practices are in this respect broader in their consequences than the questionable research practices that have dominated reform, because they compromise not merely the analysis of the data but the data themselves.

The Standards for Educational and Psychological Testing, revised in 2014 by the American Educational Research Association, the American Psychological Association, and the National Council on Measurement in Education ([American Educational Research Association, et al., 2014](#)), provide the most comprehensive framework for evaluating validity evidence, organised around test content, response processes, internal structure, relations to other variables, and consequences of testing. The revised Journal Article Reporting Standards codified expectations for transparent reporting of measures in quantitative research ([Appelbaum et al., 2018](#)), and more recent guidance has continued to systematise the reporting of quantitative

analysis at both the planning and results stages ([Schubert et al., 2025](#)). The existence of such standards, however, guarantees nothing about their uptake. A guideline can be published without being read, cited without being followed, and endorsed in principle while ignored in practice. Whether the reporting behaviour of working psychologists shifted in the wake of these frameworks is a question that has gone largely unexamined.

Measurement and the credibility revolution

The stakes of this question are raised by the direct link between measurement quality and the replicability of psychological findings. Poor measurement undermines replication through a specific mechanism: the attenuation of effects by measurement error. When an instrument lacks precision, the signal-to-noise ratio of a study falls, statistical power declines, and the probability of failing to detect a genuine effect rises across independent studies ([Loken & Gelman, 2017](#)). Hedge et al. (2017) demonstrated formally that measures with poor reliability produce highly variable effect-size estimates, rendering successful replication a near-impossibility even when the underlying effect is real. Inadequate reporting compounds this at the level of the literature, because a field that cannot observe the reliability of its measures cannot diagnose measurement error as a source of its replication failures: the information required to explain non-replication is the information most often withheld.

The replication crisis, seen in the Open Science Collaboration (2015)'s finding that only 36% of 100 published psychological studies replicated successfully, prompted a wave of methodological reform whose scope is difficult to overstate. Preregistration, registered reports, open data, open materials, and the Transparency and Openness Promotion guidelines were adopted across journals and subfields within a few years. Korbmacher et al. (2023) characterised this period not as a collapse but as a credibility revolution, marked by structural, procedural, and

community-driven improvements in research practice. If this revolution reshaped the field's methodological culture, its influence might reasonably be expected to have reached measurement practice, tightening the reporting of the instruments on which quantitative psychology depends.

The most direct evidence on this point is not encouraging. In the study most comparable to the present one, Lawson et al. (2026) coded 1,198 questionnaire measures across 506 *Social Psychological and Personality Science* articles published between 2010 and 2020, and examined whether measurement-reporting practices changed. Most did not. Reporting of scale names, citations, item counts, and reliability coefficients remained essentially flat, and the only practice to increase substantially was the availability of study materials. The authors concluded that increased attention to measurement reporting has not necessarily translated into better reporting. The reform movement, on this evidence, may have moved materials into the open without moving the reporting of measurement quality at all.

Two features of this finding define the space the present study occupies. First, it is powerful precisely because it is temporal: unlike the cross-sectional snapshots preceding it, Lawson et al. (2026) modelled change over time and found little. Second, its scope was deliberately narrow, examining a single journal, subfield, and decade, and coding what the authors called the lower-effort details that authors can report, while noting that more demanding indicators such as factor structure and convergent and discriminant validity are unlikely to be reported when even these basic characteristics are missing. Whether that flat trajectory generalises to the field at large, across clinical, cognitive, developmental, and applied psychology, across a longer window, and across the deeper dimensions of reporting the authors left uncoded, remains an open and consequential question.

The present study

The confluence of these literatures defines a specific and unfilled gap. There is abundant cross-sectional evidence that psychometric reporting is poor, a mature set of standards prescribing what it ought to be, a far-reaching period of reform that might plausibly have improved it, and, in the single study to examine change directly, a discouraging suggestion that it has not. What is missing is a systematic, large-scale, multi-subfield examination of whether the quality of psychometric reporting for newly developed and modified measures has changed over time, and, if so, whether any change has been uniform or instead confined to particular kinds of evidence, subfields, or journal strata.

This gap is increasingly consequential, because the volume of measurement in psychology is expanding rather than contracting. The proliferation of new measures, including ad hoc scales constructed for a single study, online adaptations of existing instruments, and cross-cultural modifications, has accelerated alongside the growth of online data collection and the rising output of the field ([Anvari et al., 2025](#)). If poor reporting has persisted through the open-science era despite the ready availability of guidelines, this represents a fundamental and underappreciated threat to cumulative psychological science, in which ever more theory rests upon measurement foundations that are not adequately described. If reporting has genuinely improved, this constitutes positive evidence that reform has reached measurement practice, and may help identify which mechanisms proved most effective.

The present study therefore examines whether the reporting quality of newly developed and modified psychometric measures changed across 2002 to 2026, a window spanning both the pre-reform era and the reform period that followed the replication crisis, using a large sample of published psychology articles drawn from five subfields (clinical, applied, cognitive,

developmental, and social psychology). It extends prior work in three respects: it broadens scope well beyond the single-journal, single-subfield samples of earlier studies; it lengthens the observation window to more than two decades, from before the revision of the Standards and the onset of the credibility revolution to well after; and it codes the deeper dimensions of psychometric reporting, including scale construction, participant description, validity and reliability evidence, and dimensionality, that earlier work acknowledged but did not examine.

The study accordingly addresses one primary and two secondary research questions. The primary question asks whether the overall quality of psychometric reporting for newly developed and modified measures improved across 2002 to 2026. The first secondary question narrows the focus to the type of validity evidence reported, including dimensionality, internal consistency, convergent validity, discriminant validity, and relations with other variables, since improvement concentrated in some forms of evidence but not others would carry different implications from a uniform shift, indicating whether the field is deepening its engagement with construct validity or satisfying only its least demanding expectations. The second secondary question asks whether the development procedures used to construct measures, such as expert review, cognitive interviewing, and piloting, have changed over time, addressing whether any shift in reporting reflects genuine change in how measures are built rather than only in how they are described.

Method

Description of the articles included

The final analytic sample comprised 530 articles reporting 1,142 psychometric measures, spanning a broad range of Journal Citation Reports (JCR) categories. The most represented were multidisciplinary psychology, education and educational research, and psychiatry, followed by the clinical, social, experimental, and developmental psychology categories, with a substantial

tail of others including health care sciences, criminology, and nursing. Because journals are often classified under more than one JCR category, article counts exceed the sample total; the full distribution appears in [Table A1](#). Although the sampling frame was designed around five psychology subfields, the analysed sample is characterised here by journal category rather than the intended subfield strata.

Coders

Coding was carried out by a geographically distributed team of researchers based across multiple continents, including Europe, Australia, North America, South America, and Asia. Coders varied in career stage, ranging from master's and doctoral students to early- and mid-career researchers, with early-career researchers and postdoctoral researchers forming the largest part of the pool. Demographic information such as age and gender was not systematically recorded for individual coders and is therefore not reported here.

Materials

Data sources and eligibility criteria

Article metadata were drawn from OpenAlex, an open bibliographic database whose coverage exceeds that of Web of Science and Scopus by several orders of magnitude ([Culbert et al., 2025](#)), and which provides a documented, free application programming interface (API) supporting bulk extraction, filtering, and randomised sampling. Journal-level metrics were obtained from Clarivate's Journal Citation Reports (JCR), last updated on 15 October 2025 and indexing 5,969 journals. To avoid excluding interdisciplinary work published outside dedicated psychology outlets, journals were drawn from the JCR category groups Psychiatry/Psychology and Social Sciences, General.

Eligibility was assessed in two sequential stages. The first was embedded in the OpenAlex API call, operating on article-level metadata through structured queries, one per subfield, each identified by its Wikidata concept. To be retrieved, an article had to have been published between 2002 and 2026, possess a DOI together with an ISSN or eISSN, not have been retracted, be indexed within a subfield of psychology, be of document type "article", and be written in English. The second stage operated at the journal level: duplicate records were removed, and each article's ISSN or eISSN was cross-referenced against the compiled list of journals and their JCR metrics, with any article whose identifiers failed to match excluded. This ensured that every retained article could be linked to the journal-level covariates required for analysis.

Because the eligibility criteria identified well over one million papers, manual screening of all was infeasible. A randomised subset of 10,000 articles was therefore drawn for each of five subfields (clinical, applied, cognitive, developmental, and social), yielding a starting sample of 50,000 before the removal of duplicates and non-matching records. The minimum target for fully coded articles was approximately 1,000, conservatively mirroring comparable reporting-practice reviews ([Barry et al., 2013](#)). This paper reports the articles coded up to 8 August 2026; because the models fitted are Bayesian, no statistical issue arises from analysing the data before the planned sample size is reached.

Design

Each eligible article was coded against a structured scheme, and its coded responses were converted into quantitative scores. Data were organised at three hierarchical levels. Journal-level data comprised the publisher, the Journal Impact Factor (JIF), and the cited half-life. Article-

level data comprised the year of publication, the digital object identifier (DOI), and open access status. Measure-level data comprised characteristics of each instrument, including whether it was single-item or multi-item, whether it was newly developed or a modification of an existing measure, and the types of validity and reliability evidence reported. The temporal window of 2002 to 2026 was chosen to span both the pre-reform period and the substantive methodological reform that followed the replication crisis, so that reporting practices before and after this shift could be compared within a single sampling frame.

Procedure

Screening and coder allocation

Although most filtering was embedded within the API call, coders applied two further exclusions at the point of coding. An article was excluded if it did not contain any newly generated or modified psychometric measure, or if it explicitly stated that all relevant measures were not intended for research use. Because coders worked from full-text articles, bibliographic fields such as author names, titles, and abstracts could not be obscured. To limit the bias this introduces, any coder who reported having been involved in the conception, design, analysis, or writing of an assigned article, or who reported any other conflict of interest, was reallocated to a different, randomly selected article. The screening and selection process is summarised in [Figure A1](#), reported following an adapted PRISMA 2020 flow ([Page et al., 2021](#)).

Coding scheme and outcome measures

Each article was coded against a structured scheme in which items and sub-items were scored 1 when the relevant information was reported and 0 when it was not, with no distinction drawn between the main text and supplementary materials. Free-text "other" responses were

scored 1 where one or more entries were provided and 0 where the field was blank. Coded responses were aggregated into five composite scores, whose structure is shown in [Figure A2](#).

The first composite, scale construction, drew on the items describing how a scale was developed. Multi-response items were scored one point per option selected, and the item recording whether scale items were reproduced was scored on a three-point scale reflecting whether all, example, or no items were provided; this composite ranged from 0 to 14. The second, participant description, aggregated information about the development sample and missing data, treating sample location, administration, and mode as separate sub-items, and ranged from 0 to 8. The third and fourth composites captured validity and reliability: validity ranged from 0 to 2, reflecting whether convergent and discriminant evidence were reported, and reliability from 0 to 4, reflecting the number of distinct reliability indices reported. The fifth, dimensionality, was scored separately for exploratory factor analysis or principal components analysis (0 to 10) and for confirmatory factor analysis (0 to 6), so that articles reporting either or both approaches received the appropriate score.

The primary outcome was the proportion of plausibly reportable information reported, formed by summing the composite scores and dividing by the maximum attainable for that article, which depended on the analytic approaches it adopted. Expressing the outcome as a proportion ensured that articles were evaluated only against the information it was reasonable to expect them to provide.

Intercoder reliability

Reliability of coding was assessed at three points. Before data collection, all coders independently coded a common set of 10 calibration articles, and reliability was reassessed at the midpoint and end of collection using three pre-screened articles each time. Two indices were computed: Krippendorff's alpha and simple percent agreement. Both were reported because Krippendorff's alpha can return conservatively low estimates when category usage is heavily skewed, a situation in which percent agreement provides useful context. Acceptable reliability was defined as an alpha of at least 0.67 together with percent agreement of at least 75% (Krippendorff, 2019).

During calibration, a small number of coders were directed back to the guidance documents after initial confusion about which measures to code. In these cases they had misunderstood the scope of the task: already-validated measures used without modification were not to be coded, nor were basic demographic questions such as a participant's age or education level. Once clarified, coding continued in line with the agreed scheme.

Analytic plan

The study was designed to answer three research questions. The primary question was whether the overall quality of psychometric reporting for newly developed and modified measures improved over time between 2002 and 2026. The second was whether the type of validity evidence reported, such as dimensionality, internal consistency, convergent validity, discriminant validity, and relations with other variables, has changed over time. The third was whether specific item development procedures, such as expert review, cognitive interviewing, and piloting, have changed over time. The present study's design and planned analysis was pre-registered in Open Science framework

(https://osf.io/q9xz6/overview?view_only=3b2ae45cb6d24f60a47ac2a244d10e00)

Research Question 1

The primary outcome was the proportion of applicable coding-scheme items reported for a given measure, a continuous quantity bounded between 0 and 1. Because a single article can report more than one newly developed or modified measure, an article may contribute several such proportions, which are therefore not independent of one another. To accommodate this structure, the outcome was modelled using a multilevel Bayesian beta regression (Ferrari & Cribari-Neto, 2004; Smithson & Verkuilen, 2006), which is appropriate for continuous proportional data and thus avoids the distortions that arise when bounded outcomes are treated as though they were unbounded and normally distributed, and, through a random intercept for article it accounts for the nesting of measures within articles. To capture change over time while allowing the trajectory to differ before and after the open science reforms that consolidated around 2014 (Open Science Collaboration, 2015), the fixed part of the linear predictor took a piecewise form with a breakpoint at that year:

$$g(\mu_{ij}) = \ln\left(\frac{\mu_{ij}}{1 - \mu_{ij}}\right) = \eta_{ij}$$

$$\eta_{ij} = [\beta_0 + \beta_1 \text{time}_i + \beta_2 (\text{time}_i - 12) \times \text{era}_i] + [\mu_{0j} + \varepsilon_{ij}]$$

Here, the subscripts i and j denote an individual measure and the article in which it appears, so that measures (level one) are nested within articles (level two). Time was coded so that 2002 was year zero and each subsequent value represented the number of years elapsed since 2002, and era was a dummy variable coded 0 for the pre-2014 period and 1 for the post-2014 period. Within the fixed part of the model, β_0 is the mean reporting proportion in 2002 in log-odds units, β_1 is the slope of reporting quality across the pre-2014 period, $\beta_1 + \beta_2$ is the slope across the post-2014 period, and β_2 is therefore the difference in slope after 2014 relative

to before it, with the two segments meeting at the 2014 breakpoint so that the trend remains continuous. The random part comprises two terms: μ_{0j} is a by-article random intercept, capturing the degree to which the baseline reporting proportion of article j departs from the overall average and thereby accounting for the non-independence introduced when an article contributes several measures, and ε_{ij} is the level-one residual for measure i within article j . The hypotheses for this question were directional: reporting quality was expected to be broadly flat before 2014, so β_1 was predicted to lie close to zero, and to improve thereafter, so β_2 was predicted to be positive.

Priors were specified on the log-odds scale on which the coefficients are expressed. Considering the plausible range of annual change in a proportion bounded on the unit interval, together with the expectation that reporting should not decline, the slope coefficients were assigned a $N(0.5, 0.25)$ prior, which concentrates prior mass over the plausible positive range while permitting modest decreases. The intercept, representing the expected reporting proportion in 2002, was assigned a $N(0, 3)$ prior, which comfortably spans base rates from roughly 0.1 to 0.9 in log-odds units, and default weakly informative priors were retained for the dispersion parameter. Because the model includes a random intercept for article, the between-article variance of these intercepts ($\sigma_{\mu_0}^2$) must also be estimated as we held no substantive hypothesis or prior information about the magnitude of between-article variability, this parameter was assigned a noninformative prior restricted to positive values.

Research Questions 2 and 3

For each category of validity evidence and each item development procedure, the proportion of articles reporting that category or procedure was summarised per year and per era, and temporal trends were visualised using grouped time series plots. In addition, the piecewise

beta-regression model was refitted separately to each sub-dimension of the coding scheme, namely descriptive information about scale construction, participant descriptions, validity and reliability combined, and dimensionality, with articles excluded from the dimensionality model where the relevant factor-analytic items were not reported. Because no directional predictions were made for these sub-dimensions, their slope coefficients were assigned weaker $N(0,1)$ priors.

All models were fitted in R version 4.5.2 ([R Core Team, 2025](#)) within a Bayesian framework using the brms interface to Stan ([Bürkner, 2017](#)). Posterior distributions were summarised using point estimates and 95% credible intervals, and convergence was assessed through the potential scale reduction factor and effective sample size for all parameters.

Ethics

The study analysed published, publicly accessible bibliographic and article data and did not require review by a research ethics committee. All source articles were obtained through legitimate institutional and open-access routes, and no personal or identifying data were collected. In keeping with open and reproducible research practice, the study's design, hypotheses, and analysis plan were pre-registered, and the coding materials and analysis code will be made openly available. The plan was specified before the analysis but, owing to a technical error, was not successfully submitted until afterwards, making the registration retrospective in submission; the analysis follows that plan without alteration.

Results

The final analytic sample comprised 1,142 psychometric measures nested within 530 articles. Every coded article matched its journal and publication metadata, and none fell outside the 2002 to 2026 study window. Across the full sample, the overall proportion of reporting

criteria met was low. On average, measures met 41% of the coded criteria ($SD = 0.11$), with scores ranging from 0.03 to 0.75. No measure in the sample met more than three quarters of the applicable reporting criteria. Reporting differed substantially across the four sub-dimensions. Sample description was the most completely reported, with measures meeting on average 73% of its criteria ($SD = 0.20$). Dimensionality was reported at an intermediate level ($M = 0.46$, $SD = 0.22$), computed only for the 248 measures that reported a factor analytic procedure, while scale construction was reported at a similar level to the overall score ($M = 0.37$, $SD = 0.15$). Validity and reliability evidence was the least reported by a considerable margin, with measures meeting on average only 13% of its criteria ($SD = 0.14$). Full descriptive statistics for the overall score and the four sub-dimensions are presented in [Table A2](#). The clustering of measures within articles is illustrated in [Figure A3](#), which shows the number of measures contributed by each article. While most articles reported a single measure, a substantial number reported two or more, and a small number reported many. This distribution confirmed that treating measures as independent observations would misrepresent the structure of the data, motivating the inclusion of a random intercept for article in all models.

Convergence and fit were then assessed for the primary model. All coefficients returned R-hat values at or below 1.003, indicating that the four chains had converged to a common posterior distribution, and bulk effective sample sizes exceeded 2,100 for every parameter, well above the threshold recommended for stable estimation of posterior summaries ([Vehtari et al., 2021](#)). Visual inspection of the trace plots confirmed this, showing four dense, overlapping, and stationary chains for each parameter with no drift or divergence. A posterior predictive check (as seen in [Figure A6](#) and [Figure A7](#)) indicated close correspondence between the observed distribution of reporting proportions and the distributions simulated from the fitted model,

suggesting that the beta likelihood captured the shape of the outcome adequately across the range of the scale. The four secondary sub-dimension models were subjected to the same checks and likewise converged satisfactorily, with R-hat values at or below 1.008 and effective sample sizes exceeding 1,000 for every coefficient.

Parameters (as seen in [Table A5](#)) were summarised using the median of the posterior distribution, the 95% highest density credible interval, the probability of direction as an index of the existence of an effect, and the proportion of the posterior falling within a region of practical equivalence to zero ([Makowski et al., 2019](#)).

Primary Analysis

The primary research question asked whether the overall quality of psychometric reporting changed across the study period. Neither temporal parameter reached credibility. The pre-2014 slope was negligible ($\beta = 0.005$, 95% *CrI* $[-0.010, 0.019]$), with a probability of direction of 72.62% and the entire credible interval falling within the region of practical equivalence, indicating no credible trend before 2014. The change in slope from 2014 onward, which constituted the direct test of whether reporting improved following the reporting standards revision and the reform movement, was likewise negligible ($\beta = -0.001$, 95% *CrI* $[-0.024, 0.021]$). Its probability of direction was 53.54%, barely distinguishable from chance, and its credible interval also fell entirely within the region of practical equivalence. The directional hypothesis that the post-2014 slope change exceeded zero was not supported (evidence ratio = 0.87), and the hypothesis that the post-2014 slope itself was positive was similarly unsupported. Because both credible intervals crossed zero and lay wholly within the region of practical equivalence, the finding is best interpreted not as an absence of evidence for change, but as

positive evidence for the absence of change (Kruschke (2018)). Full parameter estimates are reported in [Table A3](#) , and the modelled trend is shown in [Figure A4](#) .

Secondary Analysis

The first secondary question asked whether the reporting of specific types of validity evidence changed over time. Neither the validity and reliability sub-dimension nor the dimensionality sub-dimension showed a credible temporal effect. Validity and reliability reporting was flat before 2014 ($\beta = -0.011$, 95% *CrI* [-0.035 , 0.013]). Its post-2014 change in slope carried the highest probability of direction of any coefficient in the analysis ($\beta = 0.030$, 95% *CrI* [-0.005 , 0.066], $pd = 95.13\%$) and leaned positive, but it did not reach the threshold for a credible effect and its interval remained within the region of practical equivalence. This may be read as a faint and non-credible suggestion that reporting of validity and reliability evidence rose modestly after 2014, from the lowest baseline of any sub-dimension, but no confident conclusion is warranted. Dimensionality reporting, estimated on the subset of measures reporting a factor analysis, was also flat in both eras (pre-2014 $\beta = -0.020$, 95% *CrI* [-0.079 , 0.040]; post-2014 $\beta = 0.026$, 95% *CrI* [-0.063 , 0.116]).

The second secondary question asked whether the reporting of scale construction and development changed over time. Scale construction showed no credible change in either era (pre-2014 $\beta = 0.008$, 95% *CrI* [-0.015 , 0.029]; post-2014 $\beta = -0.014$, 95% *CrI* [-0.046 , 0.020]), and neither did sample description (pre-2014 $\beta = -0.008$, 95% *CrI* [-0.038 , 0.021]; post-2014 $\beta = 0.022$, 95% *CrI* [-0.023 , 0.068]). In every case the credible interval crossed zero and fell within the region of practical equivalence. Parameter estimates for all four sub-dimension models are reported in [Table A4](#) , and their trajectories across the study period are shown in [Figure A5](#) .

Discussion

The present study addressed whether the quality of psychometric reporting for newly developed and modified measures in psychology has changed over time. Across a multi-subfield sample of 1,142 measures nested within 530 articles, modelled with a Bayesian multilevel beta regression treating publication year as a piecewise function with a breakpoint fixed at 2014, the result was consistent: reporting quality did not change. The overall proportion of criteria met remained stable across the window, with no credible trend before 2014 and no credible change at the breakpoint marking the revision of the reporting standards and the consolidation of methodological reform. This held across all four sub-dimensions, none of which showed a credible temporal effect in either era. The single coefficient to approach conventional thresholds, the post-2014 change in validity and reliability reporting, leaned positive but did not reach credibility and remained within the region of practical equivalence to zero, indicating at most a faint, non-credible suggestion of increase from the lowest baseline of any sub-dimension.

The findings supported neither the primary hypothesis that overall reporting quality improved, nor either secondary hypothesis concerning change in the type of validity evidence or the reporting of development procedures. The interpretation of these null results rests on the estimation framework. Because the analysis was Bayesian, the flatness can be asserted rather than merely inferred from a failure to reject: the probability of direction for the primary post-2014 slope was close to chance, and its credible interval fell wholly within the region of practical equivalence, together supporting the positive conclusion that reporting did not change rather than the weaker claim that change escaped detection ([Kruschke, 2018](#); [Makowski et al., 2019](#)). This distinction is central, since a finding of stability is only compelling if the analysis can distinguish genuine absence from an underpowered null, which the precision of the present estimates confirms.

The beta likelihood, used in preference to a Gaussian model, further strengthens the inference. Because the outcome is a proportion bounded between zero and one, a Gaussian model would assign probability to impossible values beyond the scale and distort both the estimated trend and its uncertainty, most severely for sub-dimensions whose scores clustered near a boundary (Bürkner, 2017). This mattered most for validity and reliability, where many measures scored at the floor and a boundary-insensitive model would have misrepresented reporting in the domain where it was weakest.

The most revealing feature of the analysis, however, was the partition of variance between the temporal predictors and the article-level random intercept. Publication year explained a negligible fraction of the variance in reporting quality, whereas differences between articles explained the large majority. The dominant determinant of how well a measure is reported is therefore not the era of publication but the article in which it appears, and by extension the practices of its authors and the standards of its outlet.

Situating the finding in the measurement literature

Because no prior study has modelled psychometric reporting quality across so broad a sample or so long a window, direct comparison is limited. The most closely comparable work is Lawson et al. (2026), who tracked measurement-reporting practices across a decade in a single social and personality journal and found that most did not improve, the salient exception being the sharing of study materials. The present findings are consistent with theirs and extend them substantially, across five subfields rather than one, two decades rather than one, and the more demanding categories of psychometric evidence the earlier study identified as consequential but did not code. That two studies using different samples and operationalisations converge on the same conclusion lends it a robustness neither secures alone, and suggests that the stagnation of

measurement reporting is a general property of the field rather than a peculiarity of a particular journal or coding scheme.

The findings sit less comfortably alongside the broader narrative of a credibility revolution ([Korbmacher et al., 2023](#)), under which the field's methodological culture is held to have improved markedly since the replication crisis. Whatever the reach of that improvement in analytic transparency and data sharing, it did not extend to the reporting of the instruments on which quantitative claims depend. This is intelligible in light of how the reform movement defined its targets. The questionable research practices that drove reform were predominantly analytic, concerning the flexibility available in the treatment of data once collected ([Nagy et al., 2025](#); [Simmons et al., 2011](#)). Questionable measurement practices, by contrast, concern the flexibility available in the construction and selection of the instruments that generate the data, and although Flake and Fried ([2020](#)) argued that these are at least as consequential, they have received less attention from reformers. The blind spot is therefore structural: reform concentrated on what researchers do with data and largely bypassed whether the data mean what they are taken to mean.

Why has sustained awareness produced no change?

The more searching question these results pose is why. Deficiencies in psychological measurement have been documented and named repeatedly, from the reviews of the 1990s to the influential critiques of the past decade ([Flake et al., 2017](#); [Flake & Fried, 2020](#); [Hussey & Hughes, 2020](#)), and the relevant standards and reporting guidance have long been available ([American Educational Research Association, et al., 2014](#); [Schubert et al., 2025](#)). That such sustained awareness has not been accompanied by any measurable improvement is the central question these findings raise, and several explanations are worth considering.

The first concerns incentives. The reforms that succeeded, foremost the sharing of study materials, were inexpensive, visible, and increasingly mandated, whereas the thorough reporting of validity evidence is effortful, competes for limited space, and is seldom a condition of acceptance. Lawson et al. (2026) observed that the practices which improved were the least demanding, and that the more substantial forms of evidence remained absent where they were most needed. Where reform has depended on the voluntary modification of individual behaviour, its uptake has been more uneven than where change has been driven by structural and community-level mechanisms, which have been argued to underpin the more durable improvements of the reform era (Korbmacher et al., 2023).

The second locates the problem upstream: measures may frequently be used without rigorous validation, so that improved reporting cannot follow because there is nothing to disclose. This draws indirect support from evidence that shortcomings persist even when researchers are observed working rather than writing. Sladekova et al. (2026) , examining how 188 researchers conducted two analytic tasks, found that checks for basic modelling assumptions were routinely omitted or performed in ways incapable of detecting the problems present, with only a modest advantage among those who taught research methods. Sparse reporting is often a faithful record of sparse underlying practice, and remedying it would require changing what researchers do rather than only what they declare.

A third and deeper explanation situates measurement within psychology's broader difficulties with theory. Eronen and Bringmann (2021) argued that the field's recurrent crises stem partly from the difficulty of specifying and validating the constructs its theories invoke, since vaguely defined constructs cannot be measured well however carefully reporting is done. On this view, improved reporting is necessary but not sufficient, since it cannot compensate for

constructs whose theoretical specification is too weak to support valid measurement in the first place (Flores-Kanter, 2026). The stability observed here may therefore reflect not merely institutional inertia but a problem that reporting reform alone is structurally unable to solve.

Recommendations for practice

The findings carry implications for editorial practice and the culture of measurement in psychology. If two decades of awareness and reform have failed to move measurement reporting, the voluntary and educational mechanisms relied upon to date are insufficient, and structural approaches are indicated. The single reform that succeeded in the comparable literature did so because it became an expectation enforced at submission (Lawson et al., 2026). Echoing this, the present findings recommend that journals adopt a mandatory measurement-reporting checklist, verified during peer review, requiring authors either to state the validity and reliability evidence for each measure or explicitly to declare its absence. The declaration of absence is itself valuable, converting a silent omission into a visible one that readers and reviewers can weigh. Because the variance decomposition located the determinants of reporting quality at the level of articles and outlets rather than the field as a whole, editorial enforcement is well targeted: a problem residing in the practices of particular authors and journals is more amenable to gatekeeping than one diffusing across the discipline.

A second recommendation concerns the point of intervention. If sparse reporting reflects sparse practice, then reporting mandates alone will prove insufficient, since they cannot summon evidence that was never generated. Intervention earlier in the research lifecycle is therefore warranted, at the stage of measure selection and construction, alongside infrastructure that renders rigorous validation less costly. Validated measure repositories, registered measurement-development reports analogous to registered reports, and normative expectations that new

measures undergo and document formal validation before use would each address the problem at its source (Clark & Watson, 2019) . Given the proliferation of ad hoc and modified measures evident across the wide range of journal categories in the present sample, such infrastructure is likely to become more necessary over time (Anvari et al., 2025) .

Limitations and future directions

A key limitation concerns the interpretation of reporting quality as a proportion of a fixed, preregistered criterion set. Although the maxima were specified a priori to represent what could feasibly be reported, the results are conditional on that set, and the upper region of the scale may rarely have been approached in practice, since no measure exceeded three quarters of the criteria. This is compatible both with genuinely incomplete reporting and with a scheme whose ceiling was seldom attainable. Future research could validate the criterion set against exemplar measures known to be well reported, establishing whether the ceiling is reachable and calibrating the scheme accordingly.

A related limitation arises from the conditional structure of the dimensionality items, several of which were displayed only when a factor analysis extracting more than one factor was reported. Measures reporting a single-factor solution could not satisfy these criteria, and treating the unshown items as unmet may have understated their reporting quality. Future coding schemes would benefit from scoring conditional items against a denominator restricted to those applicable to each measure.

A third and more fundamental limitation concerns the distinction between the transparency of reporting and the substantive quality of the measurement reported. The outcome quantifies whether information was reported, not whether the evidence, where present, was adequate: a measure could report a factor analysis, a reliability coefficient, and validity evidence,

and thereby score highly, while those analyses were poorly conducted. The study therefore speaks to reporting transparency rather than measurement quality, and the two must not be conflated. Future research could assess the substantive adequacy of reported evidence rather than its mere presence, separating the question of whether measurement itself has improved from whether its reporting has.

A final limitation is inferential. The piecewise model tests for a change in the rate of improvement at 2014, a specific assumption about the shape of any temporal effect. A reform influence that accumulated gradually, operated with a delay, or altered the variance rather than the mean of reporting might not appear as the slope change the model was designed to detect. Two features of the results ease this concern: no trend emerged in either period, so the stability holds across the whole window rather than only around the breakpoint, and the credible intervals for both slopes were narrow and fell within the region of practical equivalence, limiting any undetected effect to a negligible size. Future work could extend this with a change-point model that estimates the location of any shift rather than fixing it.

Conclusion

The present study has addressed a significant gap in the literature concerning whether psychometric reporting quality for newly developed and modified measures has changed over time. A Bayesian multilevel beta regression indicated no credible change in reporting quality across the period 2002 to 2026, either overall or within any of its four sub-dimensions, and no detectable improvement at the breakpoint marking the standards revision and the reform period. Reporting was uniformly low, and its baseline differed markedly across sub-dimensions while remaining stable within each over two decades. The variance decomposition further indicated that reporting quality was determined more by the article and its outlet than by the era of

publication. These findings have strong implications for editorial and research practice, indicating that the methodological reforms of the past decade, whatever their success elsewhere, did not reach the reporting of measurement, and that the awareness-based mechanisms relied upon to date have proved insufficient. It is our recommendation that structural interventions, in the form of enforced reporting requirements at submission and infrastructure supporting validation earlier in the research process, are now required. Given how fundamental sound measurement is to the validity of psychological findings, it is our hope that the present study will provide the impetus for future research to examine these interventions directly, and to distinguish the reporting of measurement quality from its substance, so that the field's understanding of its most basic practices can at last begin to improve.

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Appendix

Figure A1

Screening and selection of articles, following an adapted PRISMA 2020 flow.

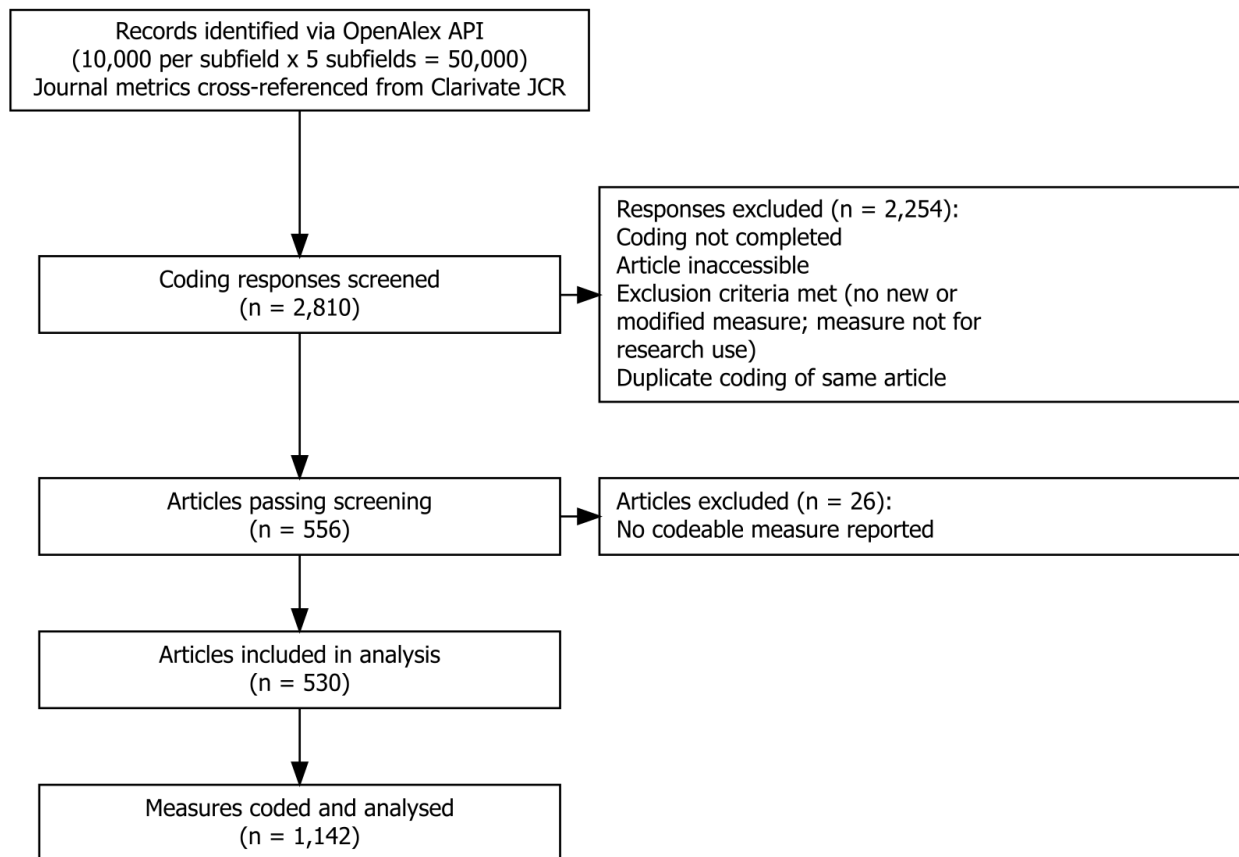


Figure A2

Structure of the coding scheme and its scoring.

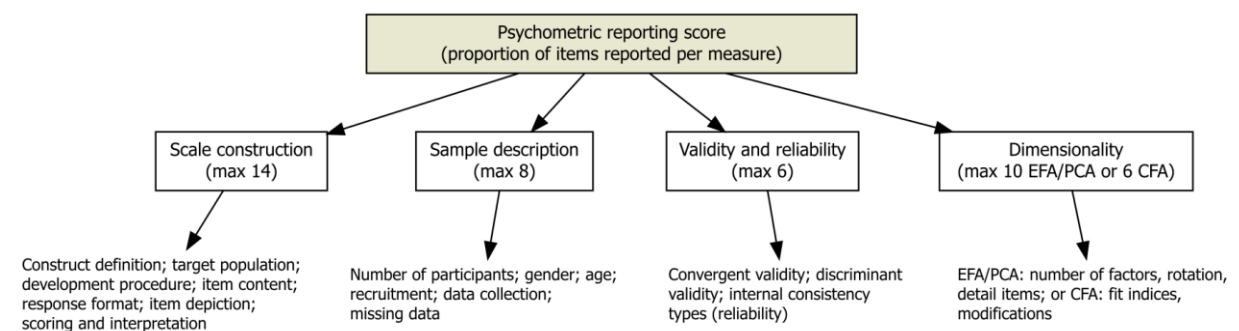


Table A1*Distribution of Analysed Articles Across Journal Citation Reports Categories*

category	n_articles
Psychology, Multidisciplinary	64
Education & Educational Research	52
Psychiatry	52
Psychology, Clinical	37
Psychology, Social	35
Psychology, Experimental	32
Psychology, Developmental	29
Psychology	27
Health Care Sciences & Services	18
Psychology, Applied	18
Psychology, Educational	18
Criminology & Penology	17
Nursing	17
Education, Special	11
Hospitality, Leisure, Sport & Tourism	11
Behavioral Sciences	10
Social Work	10
Communication	9
Social Sciences, Interdisciplinary	9
Sport Sciences	8
Education, Scientific Disciplines	5
Gerontology	4
Health Policy & Services	4
Environmental Studies	3

category	n_articles
Ethnic Studies	3
Industrial Relations & Labor	3
Law	3
Political Science	3
Sociology	3
Substance Abuse	3
Ethics	2
Social Issues	2
Transportation	2
Womens Studies	2
Economics	1
Geography	1
Public Administration	1
Social Sciences, Mathematical Methods	1

Table A2

Descriptive statistics for the overall reporting proportion and the four sub-dimensions

Variable	Mean	SD	IQR	Range	n	n_Missing
prop_score	0.41	0.11	0.13	(0.031, 0.750)	1142	0
prop_construction	0.37	0.15	0.18	(0.000, 0.882)	1142	0
prop_participants	0.73	0.20	0.25	(0.125, 1.000)	1142	0
prop_valrel	0.13	0.14	0.14	(0.000, 0.714)	1142	0
prop_dimensionality	0.46	0.22	0.44	(0.000, 0.889)	248	894

Figure A3

Distribution of the number of measures reported per article

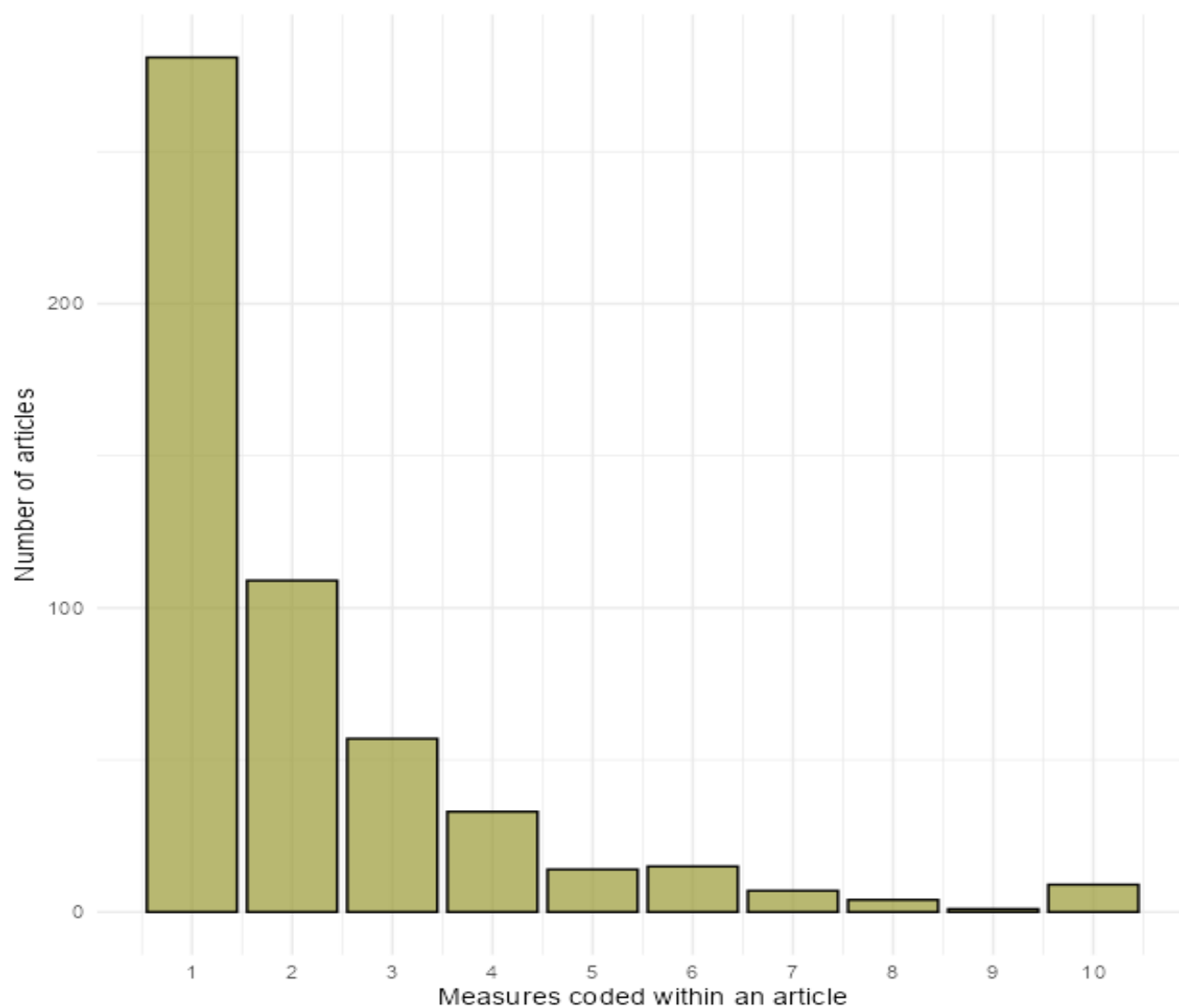


Table A3*Parameter estimates for the primary Bayesian multilevel beta model*

Parameter	Effects	Median	CI	CI_low	CI_high	pd	ROPE _Perce ntage	Rhat	ESS_tail	Group
b_Intercept	fixed	0.4096 282	0.9 5	0.5481 979	0.26611 13	1.0 00 00 0	0	1.0007 53	4801	
b_time	fixed	0.0046 942	0.9 5	0.0103 944	0.01932 81	0.7 26 25 0	1	1.0020 12	4632	
b_time_post	fixed	0.0011 093	0.9 5	0.0244 545	0.02141 42	0.5 35 37 5	1	1.0026 17	4514	
sd_doi__Intercept	random	0.4235 972	0.9 5	0.3948 276	0.45565 13	1.0 00 00 0	0	1.0004 94	6761	doi

Figure A4

Modelled trend in overall reporting quality across 2002 to 2026, with the piecewise fit and 2014 breakpoint

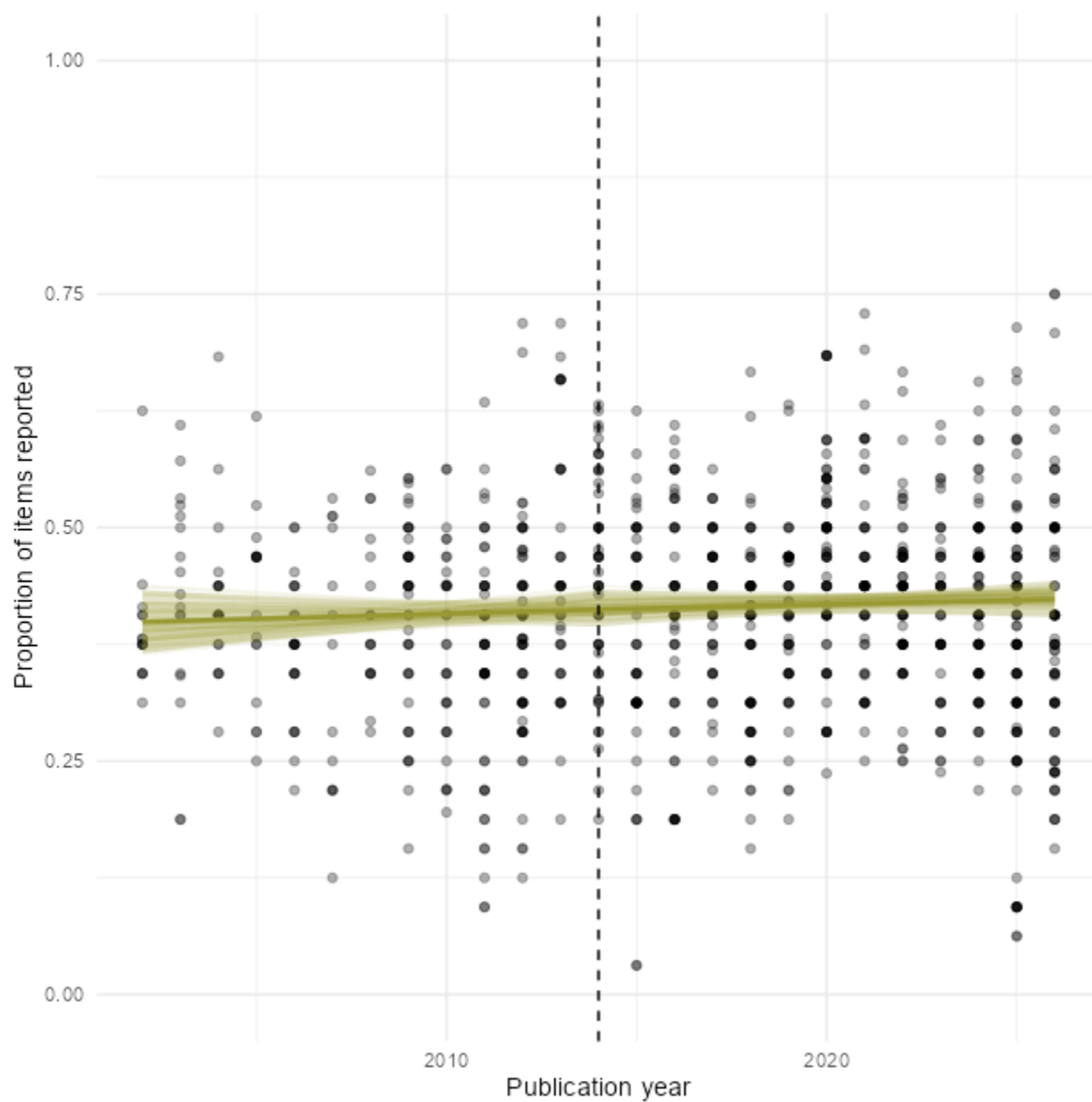


Table A4*Parameter estimates for the four sub-dimension models*

outcome	Parameter	Median	CI	CI_low	CI_high	pd	ROPE_Percentage	Rhat	ESS_tail
construction	b_Intercept	0.5184838	0.95	0.7318066	0.3253098	1.000000	0.00000000	1.000805	7791
participants	b_Intercept	0.9311792	0.95	0.6538325	1.2019594	1.000000	0.00000000	1.000683	3318
validity_rel	b_Intercept	1.4316007	0.95	1.6578821	1.2018829	1.000000	0.00000000	1.000477	5471
dimensionality	b_Intercept	0.1094024	0.95	0.4250335	0.6717177	0.646875	0.4735526	1.004114	3543

Figure A5

Reporting trajectories for the four sub-dimensions across 2002 to 2026

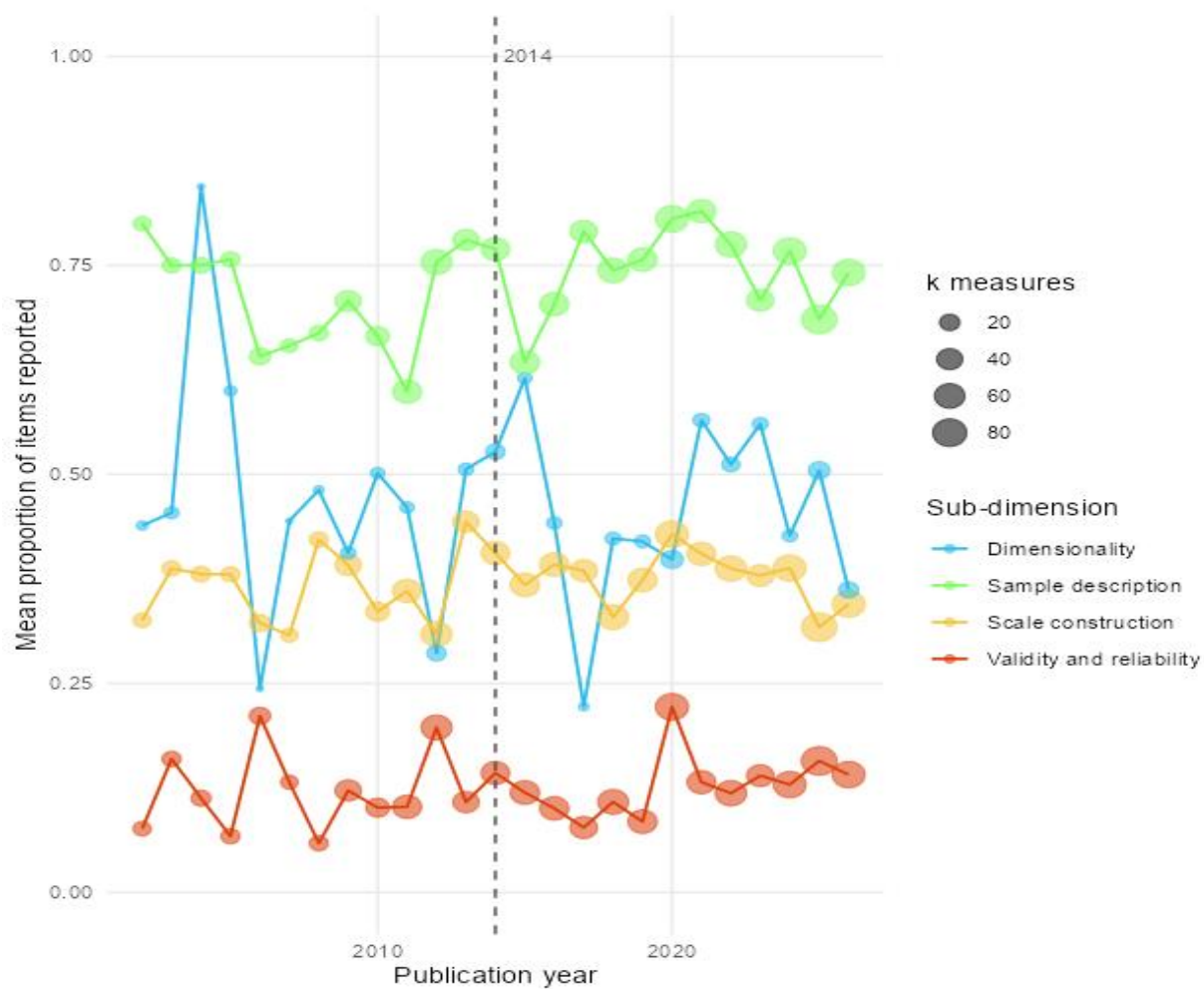
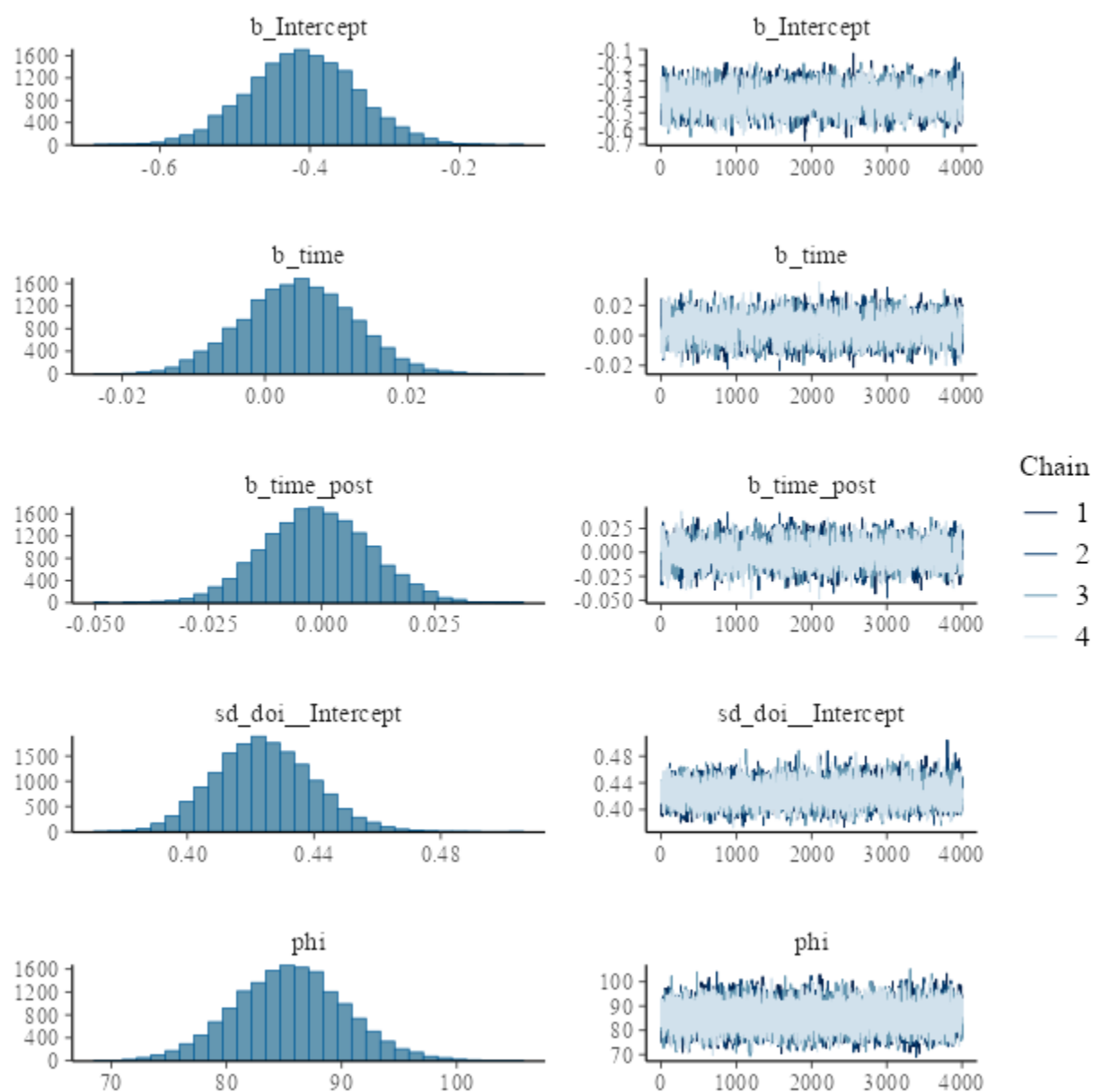


Figure A6*Trace plots and posterior predictive check for the primary model*

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Figure A7

Trace plots and posterior predictive check for the primary model

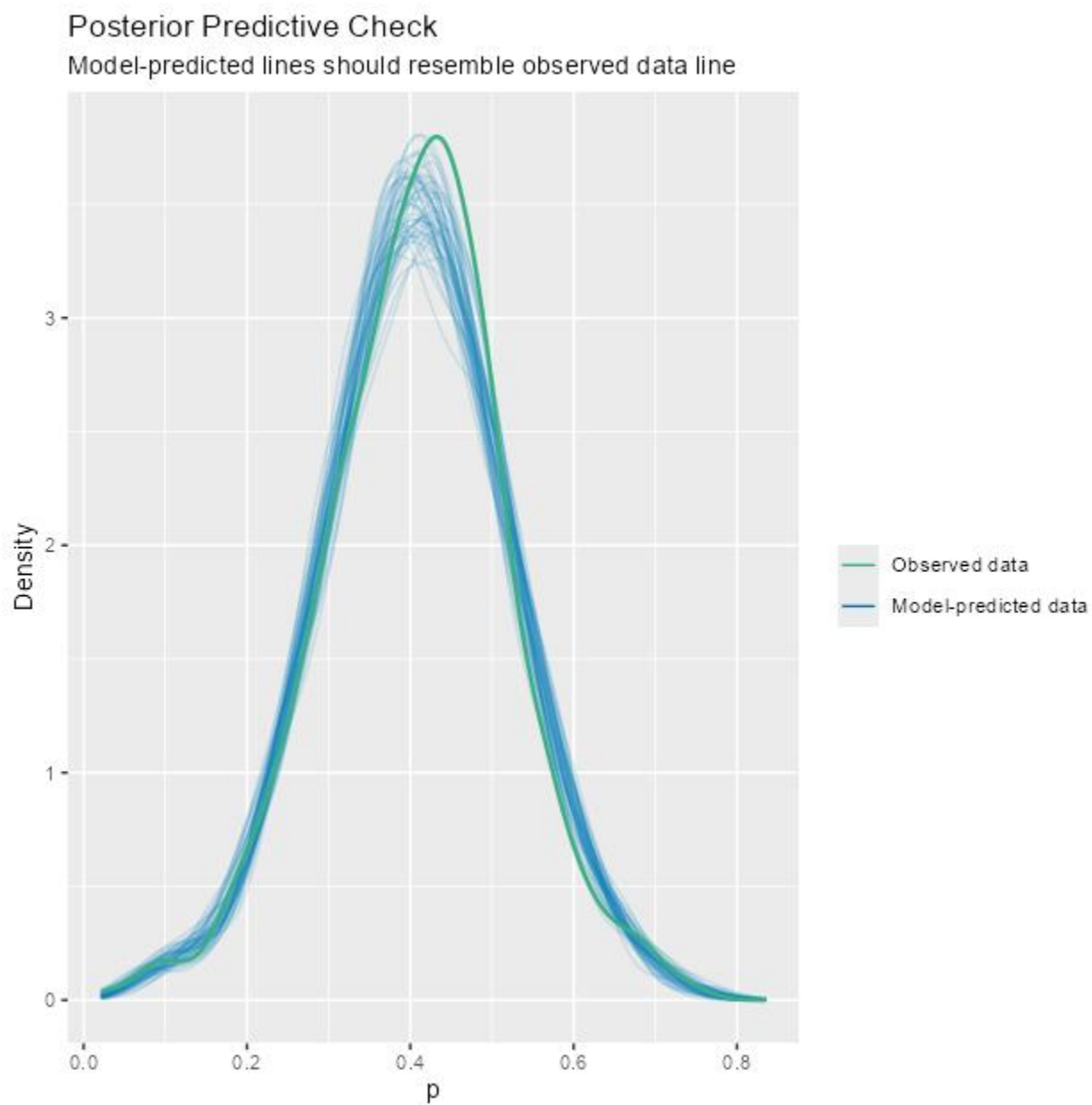


Table A5*Diagnostic Parameters of the Posterior Distribution*

Parameter	Rhat	ESS_tail	ESS_bulk	MCSE
b_Intercept	1.000753	4801	2395	0.0016388
b_time	1.002012	4632	2214	0.0001808
b_time_post	1.002617	4514	2161	0.0002897